Facial Recognition Drawer Unlock System – AI Model

Implementing CNN-based Face Verification on the XIAO ESP32-S3 Sense

System Overview

Objective: To design a secure, router-independent drawer lock actuated by live biometric facial verification.

- **Transmitter:** XIAO ESP32-S3 Sense processes live video feeds and runs Convolutional Neural Networks (CNNs) locally.
- **Receiver:** ESP32-S3 DevKit manages high-voltage (12V) relay logic to actuate the solenoid lock.
- **Communication:** Low-latency, peer-to-peer ESP-NOW protocol.

The AI Engine: ESP-WHO Framework

Bringing Computer Vision to Microcontrollers

- **What is it?** ESP-WHO is an active computer vision framework designed by Espressif specifically for resource-constrained IoT devices.
- **Architecture:** It relies on a pipeline of lightweight AI models optimized for the ESP32-S3's vector instructions.
- **Memory Utilization:** We configured the system using the **"Huge APP"** partition and allocated the heavy matrix multiplications to the board's **OPI PSRAM**.

AI Phase 1 - Face Detection & Alignment

Finding the Face in the Frame

Detection (MTCNN): The system first uses a Multi-task Cascaded Convolutional Network to scan the 1600x1200 (UXGA) frame for human facial structures.

Feature Extraction: It identifies key landmarks: the eyes, nose, and mouth corners.

Alignment: The image is mathematically cropped, scaled, and rotated so the face is standardized, removing background noise and variations in camera angle.

AI Phase 2 - Biometric Encoding

Converting Biology to Mathematics

- **The Embedding:** The aligned face is passed through a FaceNet-lite model, which strips away the visual image and outputs a highly compressed **1-Dimensional Feature Vector** (an array of numbers).
- **Verification:** To match a live face against our enrolled faces, the ESP32 calculates the Euclidean distance between the two vectors in high-dimensional space:
- **The Threshold:** If the distance, d, is below a strict mathematical threshold, the model outputs a "Match."

$$d(p, q) = \sqrt{\sum_{i=1}^{n} (p_i - q_i)^2}$$

System Initialization & Web Server Setup

Preparing the Environment

- **Wi-Fi Connection:** Upon initial boot, the XIAO ESP32- S3 Sense connects to a standard local Wi-Fi network. *Note: Wi-Fi is only required for this initial setup phase.*
- **Hosting the Portal:** The XIAO spins up a local asynchronous web server. The HTML/CSS and JavaScript for the user interface are stored directly in the C++ code as compressed hex arrays.
- **Camera Driver:** The OV2640 camera sensor is initialized via `esp_camera_init()`, piping raw JPEG frames directly to the web UI over HTTP.

The Enrollment Process

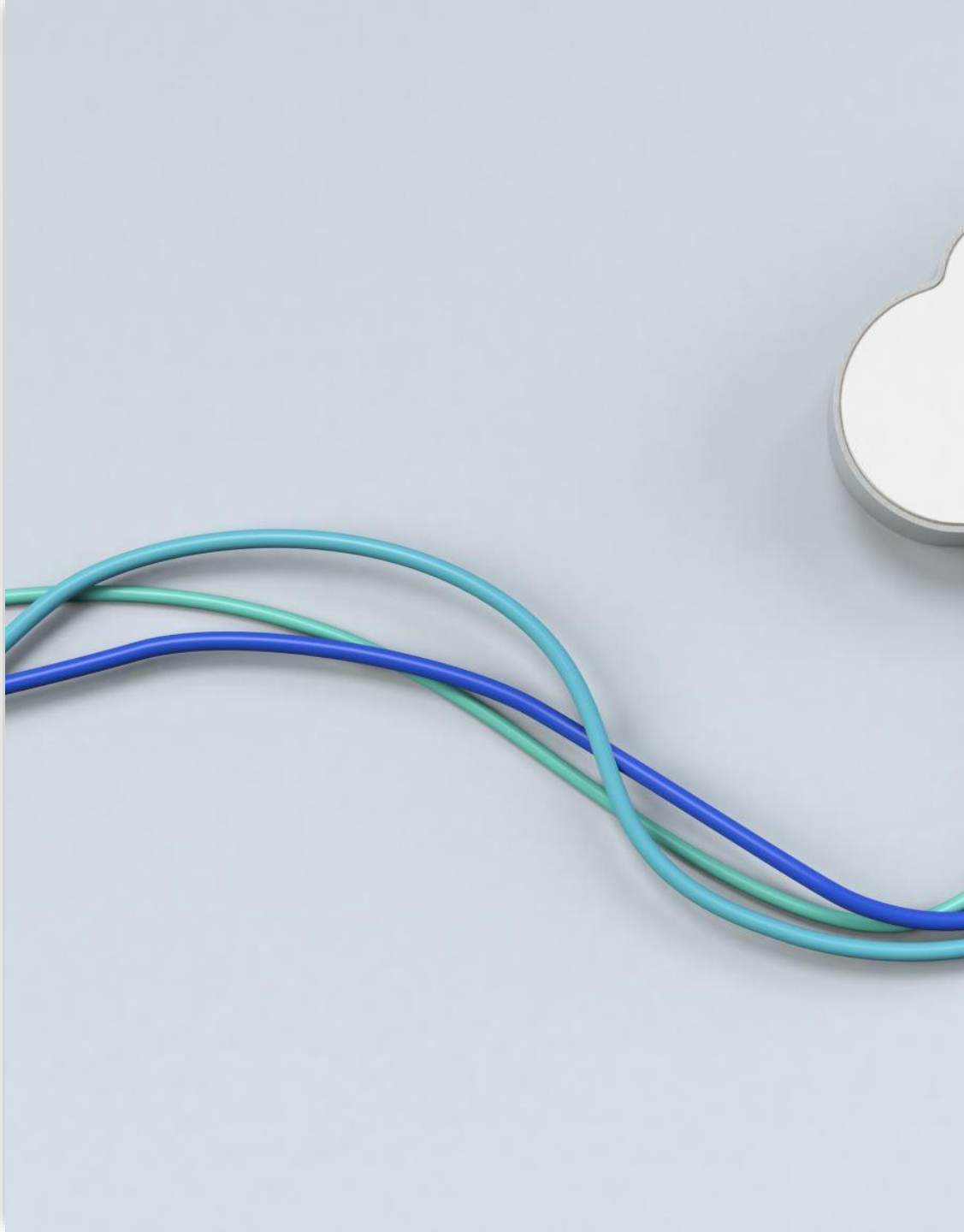
Training the Edge Device

- **Access:** The user accesses the board's local IP address via a phone or laptop browser.
- **Data Capture:** Clicking "Enroll Face" triggers the ESP-WHO model to capture 5 to 10 consecutive frames of the user's face.
- **Averaging:** The model calculates the mathematical average of these feature vectors to create a robust, single baseline profile.
- **Storage:** This profile is assigned a discrete ID (e.g., "Subject 0") and stored securely in the board's PSRAM (or Flash memory for permanent retention).

Operational Mode: Transitioning to ESP-NOW

Removing connection from the Router

- **Wi-Fi Disconnect:** Once the face is enrolled, the system no longer needs the web portal or the local Wi-Fi router.
- **ESP-NOW Activation:** The system shifts to ESP-NOW, a connectionless Wi-Fi communication protocol utilizing the 2.4GHz band.
- **The Trigger:** When a recognized face is detected, the AI script interrupts standard operation and fires a lightweight, encrypted `"UNLOCK"` payload directly to the Receiver's specific MAC Address.



Receiver Actuation & Hardware Safety

Safely Translating Logic to Physical Force

- **The Listener:** The Receiver (ESP32-S3 DevKit) constantly listens via an `onDataRecv` callback function.
- **The Translation:** Upon verifying the `"UNLOCK"` string, the DevKit drives a GPIO pin HIGH, triggering a 5V relay.
- **Hardware Protection:** The relay closes the 12V circuit to retract the lock. A 1N4007 Flyback Diode safely dissipates the inductive voltage spike when the solenoid de-energizes, protecting the microcontroller.

Conclusion

Project Summary & Future Iterations

- **Current State:** Fully functional, decentralized biometric lock operating completely on edge AI.
- **Future Upgrades:** * Migrating biometric profiles from volatile PSRAM to non-volatile SPIFFS/Flash.
- Implementing Deep Sleep interrupt triggers (via tactile button) to optimize the 3.7V LiPo battery life.

Technical Challenges

- Development Environment Constraints
- **The Roadblock:** Encountered critical configuration issues within the Arduino IDE, preventing the successful installation and compilation of facial recognition libraries.
- **Technical Context:** The required Convolutional Neural Network (CNN) models and camera drivers rely on a complex web of specific ESP32 board support packages, which proved unstable in our current IDE setup.
- **Project Impact:** This toolchain instability temporarily halted the deployment of the on-device biometric enrollment and verification logic to the Transmitter Unit (XIAO ESP32-S3 Sense).
- **Strategic Pivot:** Moving forward, the proposed solution is that we may migrate the development environment from the Arduino IDE to the official **Espressif IoT Development Framework (ESP-IDF)**, which offers native, robust dependency management for complex AI workloads.